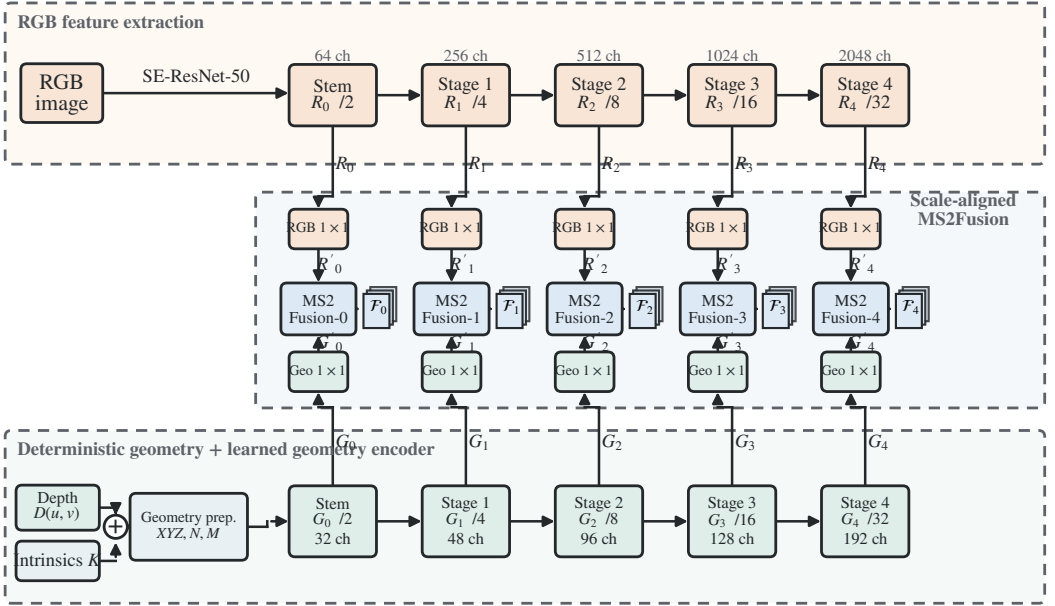
RGB feature extraction
RGB image
SE-ResNet-50
64 ch
Stem
R_0 /2
256 ch
Stage 1
R_1 /4
512 ch
Stage 2
R_2 /8
1024 ch
Stage 3
R_3 /16
2048 ch
Stage 4
R_4 /32
R_0
R_1
R_2
R_3
R_4
Scale-aligned MS2Fusion
RGB 1 × 1
R'_0
R'_1
R'_2
R'_3
R'_4
MS2 Fusion-0
MS2 Fusion-1
MS2 Fusion-2
MS2 Fusion-3
MS2 Fusion-4
$\mathcal{F}_0$
$\mathcal{F}_1$
$\mathcal{F}_2$
$\mathcal{F}_3$
$\mathcal{F}_4$
Geo 1 × 1
G'_0
G'_1
G'_2
G'_3
G'_4
G_0
G_1
G_2
G_3
G_4
Deterministic geometry + learned geometry encoder
Depth $D(u, v)$
Intrinsics K
Geometry prep. XYZ, N, M
Stem
G_0 /2
32 ch
Stage 1
G_1 /4
48 ch
Stage 2
G_2 /8
96 ch
Stage 3
G_3 /16
128 ch
Stage 4
G_4 /32
192 ch